

RESEARCH PUBLICATION (SCI/SCOPUS)

1. **C. Upadhyay**, S. S. Rajput, S. Bera, H. Ju, R. Bhide, S. Gangopadhyay, Influence of TiAlN coating thickness in dry machining of AISI 1045 steel using Finite Element simulation and experiments, *Surface and Coatings Technology*, 495, 2025, 131496. <https://doi.org/10.1016/j.surfcoat.2024.131496>
2. **C. Upadhyay**, R. Ranjan, A. Gaddam, S. Gangopadhyay, V. S. Duryodhan, Improvement in machinability characteristics of Ti6Al4V using laser surface texturing of cutting tools with varying geometry of textures, *International Journal of Advanced Manufacturing Technology*, 135, 5429–5453, 2024. <https://doi.org/10.1007/s00170-024-14795-y>
3. S.S. Rajput, **C. Upadhyay**, S. Gangopadhyay, F. Fernandes, Performance evaluation of CrAlNAg-coated inserts with varying Ag content during roughing and finishing operations in face milling. *International Journal of Advanced Manufacturing Technology*, 2024. <https://doi.org/10.1007/s00170-024-14790-3>.
4. S.S. Rajput, **C. Upadhyay**, S. Gangopadhyay, F. Fernandes, High-temperature tribological behaviour and machining performance of self-lubricant CrAlNAg coatings for dry milling operations, *Tribology International*, 198, 2024, 109824. <https://doi.org/10.1016/j.triboint.2024.109824>.
5. S. Pattanayak, S. K. Sahoo, A. K. Prajapati, A. K. Sahoo, **C. Upadhyay**, M. P. Satpathy, Adaptive control of filler wire speed in wire arc additive manufacturing: impact of inter-layer dwell time on metallurgical and mechanical aspects of ER70S-6 deposits, *International Journal of Advanced Manufacturing Technology*, 2024. <https://doi.org/10.1007/s00170-024-13981-2>.
6. A. Kumar, **C. Upadhyay**, V. Kukkala, Ch. S. Kumar, Effect of Electrode Material and Hybrid Metaheuristic Optimization on Die Corner Accuracy during Wire Electrical Discharge Machining on Inconel (625): A Comparative Study. *Journal of Materials Engineering and Performance*, 2024. <https://doi.org/10.1007/s11665-024-09649-3>.
7. **C. Upadhyay**, S. S. Rajput, Ch S. Kumar, S. Gangopadhyay, S. K. Sahoo, Performance evaluation of WC, SiAlON and SiC_w+ Al₂O₃ tools in dry machining of Inconel 617, *Journal of Manufacturing Processes*, 109, 2024, 235-249. <https://doi.org/10.1016/j.jmapro.2023.12.006>.
8. S.S. Rajput, **C. Upadhyay**, S. Gangopadhyay, F. Fernandes, Effects of roughing, finishing, and aggressive machining conditions on the milling performance of AISI 1045 steel using TiAlN coated inserts, *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture*, 2023, 095440542311579. <https://doi.org/10.1177/09544054231157963>.
9. A. Kumar, **C. Upadhyay**, N. Kumar, Soft Computing Based Parametric Optimization of Cutting Rate, Surface Roughness, and Kerf Width in Wire Electric Discharge Machining of High Strength Ti-3Al-2.5 V. *Experimental Techniques* (2023). <https://doi.org/10.1007/s40799-023-00681-x>.
10. A. Kumar, **C. Upadhyay**, Experimental investigation and optimization of machining performance characteristics during WEDM of Inconel 718: On evaluation of wire electrodes and advanced parameter techniques. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*. 2022, 236(3), 1645-1665. <https://doi.org/10.1177/09544062211023115>.
11. **C. Upadhyay**, Rahul, S. Datta, S.S. Mahapatra, B.B. Biswal, An experimental investigation on electro-discharge machining of Inconel 601, *Int. J. Ind. Syst. Eng.*, 29, 2018, 223–251. <https://doi.org/10.1504/IJISE.2018.091901>.
12. **C. Upadhyay**, S. Datta, M. Masanta, S.S. Mahapatra, An experimental investigation emphasizing surface characteristics of electro-discharge-machined Inconel 601, *J. Brazilian Soc. Mech. Sci. Eng.* 39, 2017, 3051–3066. <https://doi.org/10.1007/s40430-016-0643-2>.

13. A. Kumar, **C. Upadhyay**, R. Subbiah, D. S. Nagaraju; A hybrid approach for machining optimization of the WEDM using grey-fuzzy with Metaphor-Less algorithms for Ti-3Al-2.5V. *World Journal of Engineering*, 21(5), 2024, 894–910. <https://doi.org/10.1108/WJE-01-2023-0006>.
14. A. Kumar, , R. Subbiah, , V. Kukkala, D. S. Nagaraju **C. Upadhyay**, R. Karthikeyan, A Experimental Investigation and Soft Computing-based Assessment using ANN-MOGWO-A Hybrid Approach for Inconel(825). *Journal of Advanced Manufacturing Systems*, 23 (2), 2024; 21 (2), 305-327. <https://10.1142/S0219686724500136>.
15. A. Kumar, **C. Upadhyay**, Shashikant, Experimental investigation on WEDM performance analysis using grey-fuzzy integrated with TLBO algorithm for Inconel 625: comparison with GA and SA. *World Journal of Engineering*, 18(5), 2021, 720–733. <https://doi.org/10.1108/WJE-12-2020-0643>

INTERNATIONAL/NATIONAL CONFERENCES

1. **C. Upadhyay**, Rishabh Ranjan, S. Gangopadhyay, V.S. Duryodhan, Improvement in machinability characteristics of Ti6Al4V using laser surface texturing of cutting tools with varying geometry of textures. AdvaMAP- 2024, Lisbon, Portugal during June 10-12, 2024.
2. **C. Upadhyay**, S. S. Rajput, S. Gangopadhyay, S. K. Sahoo, Effect of tool materials, deposition technologies and cutting parameters on chip characteristics in machining of Inconel 617. AdvaMAP- 2024, Lisbon, Portugal during June 10-12, 2024
3. **C. Upadhyay**, S. S. Rajput, S. Gangopadhyay, S. K. Sahoo, Finite element simulation of chip formation and experimental characterization during dry machining of Inconel 617. AIMTDR-2023, IIT BHU, Varanasi, December 10-12, 2023.
4. **C. Upadhyay**, S. S. Rajput, S. Gangopadhyay, S. K. Sahoo, Effect of Cathodic Arc Evaporation and HiPIMS-deposited AlTiN coating in machining of Inconel 617. IECAdvaMAP- 2023, Carmel College of Engineering and Technology, Kerala during February 06-08, 2023
5. A. Kumar, **C. Upadhyay**, V. Kukkala, Study of surface characterization and parametric optimization during wire electric discharge machining for Inconel 625, *Mater. Sci. Forum*. 978 MSF (2020) 97–105.
6. T. Jadam, **C. Upadhyay**, S. Datta, S. Gangopadhyay and S. S. Mahapatra, "Analysis on topography and metallurgical aspects of EDMed work surface of Inconel 718 obtained using triangular cross sectioned copper tool electrode," *2017 International Conference on Advances in Mechanical, Industrial, Automation and Management Systems (AMIAMS)*, Allahabad, India, 2017, pp. 151-155.
7. **C. Upadhyay**, S. Datta, M. Masanta, S. S. Mahapatra, Influence of process parameters on Surface Characteristics of EDMed Inconel 601: An Experimental Investigation, *1st International Conference on Emerging Trends in Mechanical Engineering (ICETiME 2016)*, 23-24 September 2016, Faculty of Science and Technology, The ICFAI Foundation of Higher Education (IFHE), Hyderabad.
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